INCH-POUND

MIL-B-71190 (AR)
24 August 1994

MILITARY SPECIFICATION

BAG, MULTIWALL PASTED VALVE

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for one type of multiwall pasted valve bag for use in packaging bulk explosives (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1905 - Design and Test Requirements for Level B and Level C Ammunition Packaging

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12972282 - Bags, Five Ply

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

DOCUMENTS

49CFR - Code of Federal Regulations Title 49
- Transportation of Hazardous Materials

(The Code of Federal Regulations (CFR) are on sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

TAPPI-T402 Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets, and Related Products
TAPPI-T404 Tensile Breaking Strength of Paper and

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Paperboard
TAPPI-T410 Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard
TAPPI-T456 Wet Tensile Breaking Strength of Paper and Paperboard

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, Technology Park/Atlanta, P.O. Box 105113, Atlanta, GA 30348.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D3951 - Commercial Packaging, Practice for (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Multiwall pasted valve bag. The multiwall pasted valve bag shall comply with all requirements specified on drawing (dwg.) 12972282 and with all requirements specified in applicable specifications and standards.

3.2 Construction. The multiwall pasted valve bag shall have a tuck in sleeve (inner flap), an opening or valve, in one corner through which the bag is filled.

3.2.1 Plies. The bag shall have five (5) plies with a glue seam for the top and bottom and an overlap side glue seam. The innermost ply and tuck in sleeve shall be made of No. 50 natural kraft with a 0.0015 inch minimum thickness of polyethylene applied to the innerside of the ply and tuck in sleeve. The second ply shall be No. 50 kraft. The third ply shall consist of a high density polyethylene sheet stretched in one direction and a second layer of high density polyethylene bonded to it at 45 degrees. The total thickness shall be 2.5 mil minimum. No. 50 kraft shall be used for the forth and fifth plies.

3.2.2 Bonding of plies. The top and bottom ends shall be continously bonded, with hot melt or thermoplastic adhesive, so that the various walls will be securely anchored, one to another providing strong closures. Additional glue shall be sprayed into the corner seams (to control seams). One top corner shall not be pasted so as to provide a filling valve.

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3.3 Adhesive water resistance. The adhesive, applied to the seams and closures, shall be water resistant to the extent that there will be not more than 25 percent of adhesive failure of the bonded area. The adhesive shall be hot melt or thermoplastic adhesive.

3.4 Seam strength. The adhesive, applied to the seams, shall give a seam strength at least equal to the wall of the kraft paper. The seams shall have a continous bond. The overlap of paper at the longitudinal seam shall not be less than one half (1/2) inch and there shall not be more than 3/16 inch of unsealed edge on the exposed edge.

3.5 Basis weight of kraft paper. The basis weight of single ply kraft paper used in the fabrication of the bags shall be fifty (50) pounds (24 x 36 -500). The tolerance for basis weight is minus 5 percent and unlimited plus tolerance.

3.6 Leakproofness. The filled bag shall not exhibit any penetration (leakage) of the contents.

3.7 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the techncial provisions herein (see 4.3).

3.8 Workmanship. The bags shall be uniformly constructed and shall be free from holes, tears or other defects that would impair their usefulness or which might result in damage to the contents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does

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not authorize submission of known defective material, either indicated or actual nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078, apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (See Table I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Multiwall Pasted Valve Bags	1 OF 1	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Multiwall pasted valve bags destructive inspection. Examination for defects	5	3.1	4.4.2.1
	Multiwall pasted valve bags non destructive inspection. Examination for defects	10	3.1,3.2	4.4.2.2
	Packaging of bags Examination for defects	5	5.1	4.4.2.3
	Leakproof test of bags	12	3.6	4.5.1

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2. Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels			
	III	IV	V	VI
2 to 8	*	*	5	3
9 to 25	*	13	5	3
26 to 50	32	13	5	3
51 to 150	32	13	13	5
151 to 280	32	30	14	8
281 to 500	32	30	17	8
501 to 1200	74	35	20	13
1201 to 3200	74	43	24	13
3201 to 10000	87	50	30	13
10001 to 35000	109	61	36	13
35001 5o 150000	124	74	40	13
150001 to 500000	156	91	40	13
500001 and over	187	102	40	13

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Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc. may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-B-71190 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Bag, Multiwall Pasted Valve (destructive inspections)		NA
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101.	Adhesive water resistance	6, 0-1	3.3	4.5.2
102.	Seam strength	6, 0-1	3.4	4.5.3
103.	Basis weight	6, 0-1	3.5	4.5.4
104.	Incorrect number of plies	6, 0-1	3.1	Visual
105.	Innermost ply not coated or incompletely coated with polyethylene of correct thickness	6, 0-1	3.1	Visual/Certification
106.	Third ply not correct	6, 0-1	3.1	Certification
107.	Remaining plies not correct kraft paper	6, 0-1	3.1	Certification
108.	Incorrect adhesive		3.2.2	Certification
Minor				
201.	Evidence of poor workmanship	6, 0-1	3.8	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Bag, Multiwall Pasted Valve	1	1	12972282
				NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101.	Tuck in sleeve missing	Level III	3.2	Visual
102.	Seams not sealed (glued) or incompletely sealed	Level III	3.4	Visual
103.	Length (empty bag)	Level III	3.2	Scale
104.	Width (empty bag)	Level III	3.2	Scale
105.	Longitudinal seam incorrectly overlapped	Level III	3.4	Scale
106.	Width of unsealed edge on exposed edge	Level III	3.4	Scale
Minor				
201.	Evidence of poor workmanship	Level IV	3.8	Visual

NOTES:

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Packaging, Packing and Marking	1 OF 1	NA
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101.	Packing or packaging materials missing or incorrect	Level III	5	Visual
Minor 201.	Evidence of poor workmanship	Level IV	5	Visual

NOTES:

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4.4.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in Table I and the "Inspection Methods Reference" column of the Classification of Characteristics listing starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection. (see 6.5)

4.5.1 Leak test. (First article only).

4.5.1.1 Preparation. Fill each bag to capacity (42 ± 2 pounds) with an inert filler (see 6.6). The filler material shall have a bulk density of 0.997 + 0.005 g/ml. Seal the bag by securing the sleeve shut with tape and then tucking the sleeve in the bag. Pre-condition the bag in accordance with TAPPI T402. Immediately after pre-conditioning, condition the bag for twenty four (24) hours in accordance with TAPPI T402.

4.5.1.2 Procedure. Perform tests in the sequence indicated ("a" through "e") on the loaded and sealed bags. All testing shall be performed at ambient temperature. Perform the tests below, "a through d", in accordance with Table I (Packs weighing 150 pounds or less), Level C (CONUS) of MIL-STD-1905. Test "e", the four foot drop, shall be in accordance with 49 CFR, Paragraph 178.603. If any spillage or sifting of the contents of the bag is observed, after any test, the bag shall be classed defective.

a. Stacking. Divide the bags into two sets. Six bags in each set. For each set of loaded and sealed bags, use a mini-pallet (two, 3 bag layers) securely strapped with 1/2 inch plywood planks on all sides.

b. Secured vibration. Use the same mini-pallet configuration as specified above.

c. Two foot drop. Perform this test on each bag individually.

d. Loose cargo test. Perform this test using three (3) bags at a time.

e. Four foot drop test. Perform this test each bag individually.

4.5.2 Adhesion water resistance. Preconditon and condition the bags in accordance with TAPPI T402. Determine the water resistance of the adhesive of the outer ply longitudinal seam and one of the pasted end closures in accordance with TAPPI T456, except as follows:

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Cut test specimens one (1) inch wide so that the longitudinal seam or pasted end closure runs perpendicular to and is centered relative to the long dimension of the specimen. The test specimen shall encompass all adhesive bond areas included in fabricating the seam or end closure. In the case of multi-ply end closures, clamp all plies in the jaws of the tester. Immerse the specimens in not less than 1 inch of the distilled water for 24 hours. Run a wet tensile test. A test specimen fails the test if failure occurs with the separation of the seam or closure and less than 25 percent of the separated adhesive area shows fiber tear. Failure of more than 10 percent of the specimens shall be reported as failure of the adhesive.

4.5.3 Seam strength. Precondition and condition the bags in accordance with TAPPI T402. Determine the tensile strength of the glued longitudinal seam in accordance with TAPPI T404. Cut the specimen so that the seam runs perpendicular to the long dimension of the specimen. When clamped in the machine for testing, locate the seam midway plus or minus 1/2 inch between the two jaws of the tensile testing machine. If upon conclusion of the test, failure occurs with the separation of the seam, the specimen shall be acceptable if the separated adhesive area shows not less than 75 percent fiber failure rather than adhesive failure. Seam strength requirements shall be satisfactory if specimen failure occurs in areas other than seam separation.

4.5.4 Basis weight of kraft paper. Precondition and condition the samples in accordance with TAPPI T402. Perform this test in accordance with TAPPI T410.

5. PACKAGING

5.1 Preservation. Preservation shall be in accordance with ASTM-D 3951.

5.2 Packing. Packing shall be in accordance with ASTM -D3951.

5.3 Marking. Marking shall be in accordance with ASTM -D3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The multiwall pasted valve bags are intended for use in packaging finished bulk explosives. Among the explosives which will be packaged with these bags are Composition B, Composition A-5, Octol 75/25, LX-14-0 and other explosive compositions as permitted by their respective specifications.

6.2 Acquisition requirements. Acquisition documents must specifiy the following:

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a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Requirements for submission of first article sample.

d. Certificate of conformance for each shipment of product.

e. Packaging requirements, if other than specified in Section 5.

6.3 Submission of alternative quality conformance provisions. All proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-Q, Picatinny Arsenal, New Jersey 07806-5000. This description should include, but not be limited to, the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.6 Inert filler. An inert filler composition, for use in filling the bags which will be tested, is a mixture of 75 percent cellulose acetate to 25 percent sand. As an alternative, any non-hygroscopic powder or fine granular mixture having an approximate Bulk Density of 62.4 pounds per cubic foot is acceptable.

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6.7 Subject term (key word) listing.

Explosive, Finished Bulk
Kraft paper
Packaging
Polyethylene

Custodian
Army-AR

Preparing activity
Army-AR

(Project 8140-A992)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-71190 (AR)	940824

3. DOCUMENT TITLE: BAG, MULTIWALL PASTED VALVE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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